62%

INFORMATION COMMUNICATION TECHNOLOGY

Object-Oriented Programming in C++
Module 4
CAT 3

Time: 1 hour 30 minutes | **Total:** 40 marks

Candidate Name: _____

Admission Number: _____ **Date:** _____

Instructions

1. Answer ALL questions.
2. Write your name and admission number in the spaces provided above.
3. Write all code neatly and use proper C++ syntax.

QUESTIONS

1. Define **object-oriented programming** and state what an object bundles together. (2 marks)
2. Explain the difference between a **class** and an **object**, using a real-world example. (3 marks)
3. Compare **procedural programming** and **object-oriented programming** in terms of focus, handling of data, and reuse. (3 marks)
4. Explain the OOP concepts of **encapsulation** and **polymorphism**, giving one example of each. (4 marks)

5. State **two advantages** and **two disadvantages** of object-oriented programming. (4 marks)
6. State the purpose of each of the following parts of a C++ program: (4 marks)

   a) #include <iostream>

   b) using namespace std;

   c) int main()

   d) return 0;

7. The program below contains errors. Identify **four** errors and write the corrected statements. (4 marks)

```
8. #include <iostream>
9. using namespace std ;
10.
11. int main() {
12.     int age;
13.     cin << age;
14.     cout >> "Age is " >> age;
15.     return 0;
16. }
```

17. a) Describe the three C++ access specifiers: **public**, **private** and **protected**. (3 marks)

    b) A programmer declares the balance attribute of a BankAccount class as **public**. Evaluate this decision and recommend a better approach. (3 marks)

18. a) Write a C++ class named Book with a string attribute title, an int attribute price, and a method display() that prints both values. All members should be public. (3 marks)

    b) Write the statements in main() that create an object of Book, assign values to its attributes, and call the display() method. (2 marks)

19. Write a complete C++ program that defines a class Rectangle with a method calculateArea(int length, int width) that returns the area. In main(), create an object, call

the method with values of your choice, store the result in a variable, and display it. (5 marks)

1. Object oriented programming is a programming paradigm based on object that contain data and code. 8½
An object includes data, attributes and methods.

2. A class is a blueprint while object is instance of a class
A car = class while toyota = object 8½

3.

	Procedural Programming	Object oriented Programming
Focus	Focuses on functions/procedures	Focuses on objects
Data Handling	All data and methods are segregate.	All data and methods are bundle together inside objects (Encapsulation).
Reuse	Code reuse is limited.	High reusability through class inheritance and Polymorphism.

4. Encapsulation → Hiding internal details and restricting access.
Private balance in bank account with public withdraw methods. 8½
Polymorphism → Same interface but different behaviour.
Shape draw() works differently for circle and rectangle.

5. Advantages
OOP is reusable in big scale.
Have easier maintainance. 04
Disadvantages
Complex to learn
Can be more memory-intensive than procedural.

6. #include <iostream> . This include input/output library.
Using namespace std; Allows use of std names (cout, cin) without std:: Prefix.
int main() Show entry point of main Program. 04
Return 0; Indicates successfull Program termination.

17

7. Missing semicolon after std ✓ 0

using namespace std;

Cin << age; Should be Cin >> age; 1/2

Cout >> "age is" >> age; Should be Cout << "age u" << age; ✓ 1/2

Public - Accessible to anywhere in the program ✓

Private - Accessible only inside the main Program.

Protected - Accessible inside the class and derived classes ✓

4. Not best declaration decision since balance can be changed freely 0/2

It better make it Private and provide public method for controlled access.

10.

(b)

```
class book {
Public:
    string title;
    int price;
    Void displaydetails()
    { where is price?
    cout << tittle << 'Price';
    } };
```

```
Book b;
b.title = "Programming guide";
b.Price = 500;
b.displaydetails();
```

-0/2

0/9

```
#include <iostream>
using namespace std;
class Rectangle {
  public:
    int calculateArea(int width, int length)
    {
    return length * width;
    }
};
int main()
{
Rectangle r;
int area = r.calculateArea(6, 11);
Cout << area;
return 0;
}
```

05

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